Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I think loops, conditionals, dataclasses, and lists will all be helpful on this project. I'm hoping to improve my drafter skills, specifically pages and routes.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

The easiest will probably be defining my dataclasses. The hardest will probably be making sure all of my routes are called correctly and return what they're supposed to.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

The first thing I plan to implement is the ability to input schedules.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Distractions include other classes, competition prep, clubs, and books. I plan to set aside time each week to work on the project outside of those things.

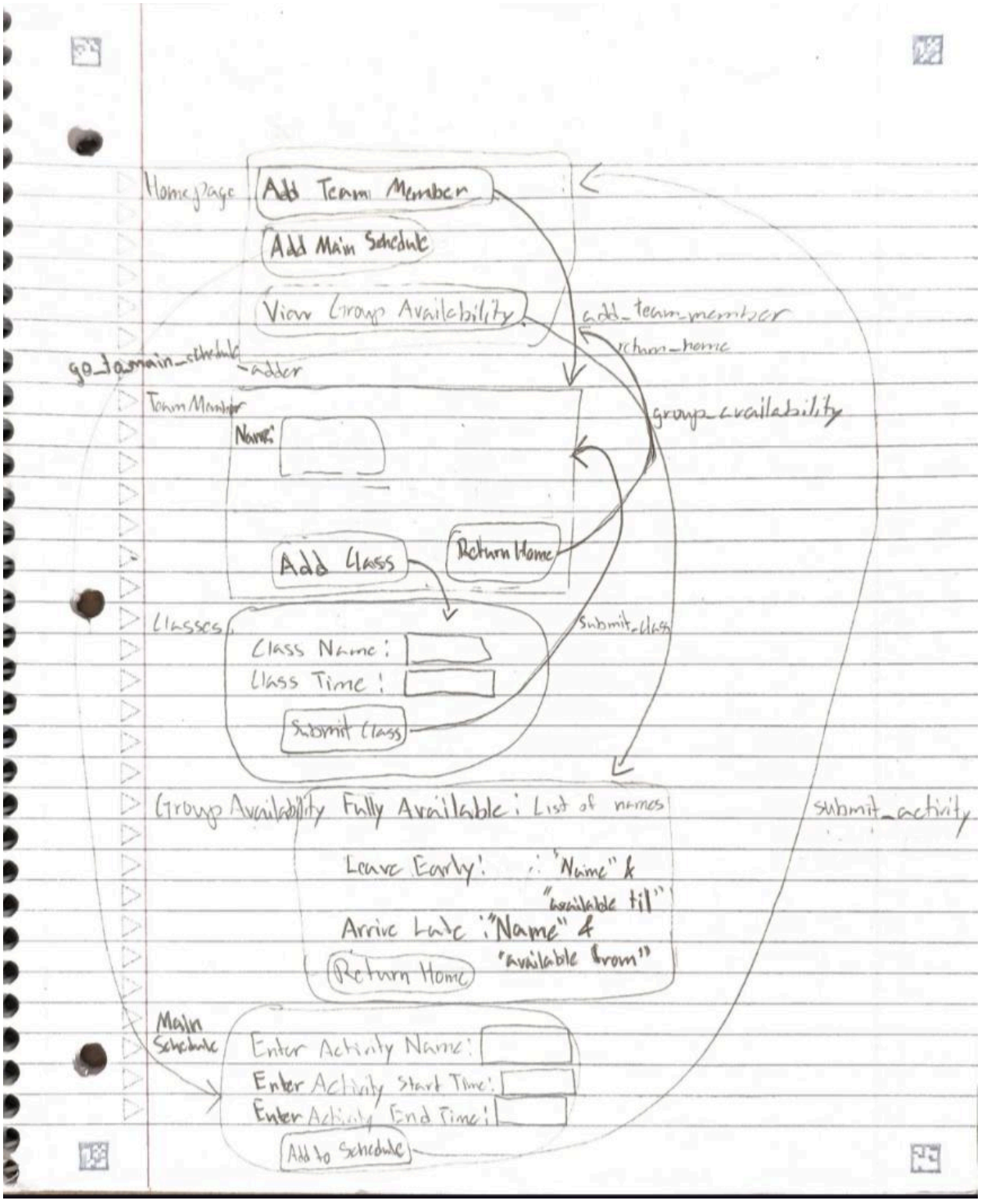
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


Homepage
Add Team Member
Add Main Schedule
View Groups Availability
add_team_member
return_home
go_to_main_schedule_adder
groups_availability
Team Member
Name:
Add Class
Return Home
Classes
Class Name:
Class Time:
Submit Class
submit_class
Groups Available: List of names
Fully Available:
Leave Early: : "Name" & "available til"
Arrive Late: "Name" & "available from"
Return Home
submit_activity
Main Schedule
Enter Activity Name:
Enter Activity Start Time:
Enter Activity End Time:
Add to Schedule

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**Make it easier for UDSST presidents to create groups for the scavenger hunts during recruitment weekends.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	Class State: Team_members: list[TeamMember] Main_schedule: list[Activity]
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Class TeamMember: Name: str Courses: list[Course] Class Course: Name: str Start_time: str End_time: str Class Activity: Name: str Start_time: str End_time: str

Routes (minimum 7) The names of your route functions	Add_course Add_teammember Group_availability Return_home Submit_course Go_to_main_schedule_adder submit_activity
Pages (minimum 5) A brief description of the pages in your site	Homepage: Hub where you start and end and navigate to many other pages. AddTeamMember: you can add TeamMembers here. Courses: Accessed through AddTeamMember, you add courses to TeamMembers here. ViewAvailability: You can see when TeamMembers are available. AddMainSchedule: Lets the president add the activities on the general schedule.
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	There are text boxes to enter classes and to enter activities and names and start/end times.
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	For user input for class time (If not time: Return “Enter valid time”) For user input for activity time (If not time: Return “Enter valid time”) For user input for name (If not name: Return “You haven’t entered a name. Please enter a

	name to continue”)
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	For course in TeamMember.courses: Availability = check_availability(course) Return availability Checks if a teammember is available which is the whole point of this program

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I have not used an LLM.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.